Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-5 — Completing the Square

Classwork — Give it a go

Show all your working. Add the adjustment term to BOTH sides. A correct answer with no working is only half an answer.

Student name :	Date : .....

Part A — The adjustment term and solving Questions 1 to 4

1. Complete the square and write the result as a squared bracket.

$$x^2 + 14x + \square$$

2. Complete the square and write the result as a squared bracket.

$$x^2 - 9x + \square$$

3. Solve by completing the square.

$$x^2 + 10x + 9 = 0$$

4. Solve by completing the square.

$$x^2 - 4x - 21 = 0$$

Part B — Coefficient not 1, and complex solutions Questions 5 to 7

5. Divide by the leading coefficient first, then solve.

$$2x^2 - 4x - 6 = 0$$

6. Solve. Write both solutions in the form $a + bi$.

$$x^2 + 2x + 10 = 0$$

7. Solve. Give the exact solutions.

$$3x^2 - 6x - 1 = 0$$

Part C — Extreme values and an error Questions 8 to 10

8. Write in the form $(x-h)^2+k$ and state the minimum value and where it occurs.

$$f(x) = x^2 - 10x + 30$$

9. Write in the form $-(x-h)^2+k$ and state the maximum value and where it occurs.

$$g(x) = -x^2 + 6x - 4$$

10. A student solves $x^2+6x=7$ like this. Say what is wrong and give the correct solutions.

$$x^2 + 6x + 9 = 7 \;\Rightarrow\; (x+3)^2 = 7$$